Table 1: **Reasoning post-training: every checkpoint of Table ??.** Before and After are the checkpoints before and after each OT3-30K update, not before and after Tulu3. Starting checkpoints are evaluated again here, so these values are paired within the table and must not be compared with the Phase 1 tables. Protocol B (Appendix Table 5). Values are mean$_{\pm SD}$ over three sampled runs, and Avg is the mean over the eleven tasks shown.

(a) Mathematics

Model	Stage	A24	A25	AMC	M500	GSM8K
Main comparison: Tulu3 SFT → OT3-30K						
MV	Before	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$8.3_{\pm 3.82}$	$29.2_{\pm 6.41}$	$46.3_{\pm 0.88}$
	After	$11.1_{\pm 3.85}$	$16.7_{\pm 3.33}$	$46.7_{\pm 6.29}$	$71.0_{\pm 1.06}$	$49.0_{\pm 1.75}$
FineWeb-Edu	Before	$0.0_{\pm 0.00}$	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$0.8_{\pm 0.53}$	$7.3_{\pm 0.76}$
	After	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$0.6_{\pm 0.00}$	$6.2_{\pm 0.72}$
Nemotron-CC-HQ	Before	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$5.0_{\pm 5.00}$	$1.9_{\pm 0.31}$	$5.5_{\pm 0.40}$
	After	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.3_{\pm 0.12}$	$2.9_{\pm 0.09}$
Comma0.1	Before	$0.0_{\pm 0.00}$	$1.1_{\pm 1.92}$	$2.5_{\pm 2.50}$	$1.7_{\pm 0.61}$	$11.0_{\pm 0.77}$
	After	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.3_{\pm 0.58}$	$13.4_{\pm 0.80}$
SmolLM2	Before	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 2.89}$	$6.7_{\pm 0.70}$	$25.9_{\pm 1.45}$
	After	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$6.6_{\pm 0.92}$	$23.1_{\pm 1.34}$
Qwen2.5	Before	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$6.7_{\pm 1.44}$	$21.0_{\pm 2.78}$	$55.1_{\pm 0.39}$
	After	$4.4_{\pm 5.09}$	$1.1_{\pm 1.92}$	$26.7_{\pm 7.22}$	$44.9_{\pm 2.00}$	$57.6_{\pm 0.34}$
Qwen3	Before	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$12.5_{\pm 2.50}$	$24.7_{\pm 2.50}$	$64.7_{\pm 0.80}$
	After	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$1.7_{\pm 1.44}$	$10.9_{\pm 0.42}$	$25.7_{\pm 0.38}$
MV starting-point controls						
16K extended	Before	$3.3_{\pm 3.33}$	$8.9_{\pm 3.85}$	$25.8_{\pm 12.58}$	$46.5_{\pm 11.29}$	$55.0_{\pm 1.78}$
	After	$8.9_{\pm 5.09}$	$25.6_{\pm 1.92}$	$42.5_{\pm 6.61}$	$69.8_{\pm 1.25}$	$58.8_{\pm 1.52}$
Tulu3 SFT + DPO	Before	$1.1_{\pm 1.92}$	$1.1_{\pm 1.92}$	$14.2_{\pm 6.29}$	$30.1_{\pm 3.31}$	$47.4_{\pm 1.20}$
	After	$0.0_{\pm 0.00}$	$4.4_{\pm 1.92}$	$17.5_{\pm 4.33}$	$34.4_{\pm 1.78}$	$44.4_{\pm 1.52}$

(b) Code, science and instruction following, and the average

Model	Stage	HE	LCB	MBPP	GPQA	JEE	IFE	Avg
Main comparison: Tulu3 SFT → OT3-30K								
MV	Before	$44.9_{\pm 1.76}$	$7.0_{\pm 2.26}$	$31.9_{\pm 0.46}$	$27.4_{\pm 0.58}$	$7.8_{\pm 1.33}$	$54.0_{\pm 0.15}$	$23.4_{\pm 0.31}$
	After	$51.0_{\pm 4.97}$	$11.5_{\pm 0.87}$	$28.5_{\pm 3.29}$	$28.8_{\pm 1.75}$	$18.8_{\pm 1.34}$	$42.1_{\pm 0.20}$	$34.1_{\pm 0.93}$
FineWeb-Edu	Before	$2.2_{\pm 1.96}$	$0.1_{\pm 0.16}$	$1.0_{\pm 0.20}$	$28.6_{\pm 1.27}$	$3.1_{\pm 0.59}$	$38.1_{\pm 0.98}$	$7.5_{\pm 0.37}$
	After	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$23.9_{\pm 0.77}$	$1.9_{\pm 0.44}$	$33.4_{\pm 0.67}$	$6.2_{\pm 0.08}$
Nemotron-CC-HQ	Before	$3.5_{\pm 0.93}$	$0.0_{\pm 0.00}$	$1.9_{\pm 0.23}$	$28.5_{\pm 1.77}$	$3.2_{\pm 0.39}$	$39.2_{\pm 1.78}$	$8.1_{\pm 0.89}$
	After	$0.2_{\pm 0.35}$	$0.0_{\pm 0.00}$	$0.0_{\pm 0.00}$	$25.1_{\pm 2.96}$	$1.1_{\pm 0.20}$	$35.5_{\pm 0.34}$	$5.9_{\pm 0.31}$
Comma0.1	Before	$14.2_{\pm 0.93}$	$0.4_{\pm 0.31}$	$16.5_{\pm 2.08}$	$26.4_{\pm 5.40}$	$5.5_{\pm 0.72}$	$46.8_{\pm 0.52}$	$11.5_{\pm 0.82}$
	After	$1.4_{\pm 0.93}$	$0.0_{\pm 0.00}$	$0.1_{\pm 0.12}$	$26.8_{\pm 2.81}$	$2.6_{\pm 0.32}$	$36.8_{\pm 0.08}$	$7.5_{\pm 0.29}$
SmolLM2	Before	$23.6_{\pm 2.46}$	$0.9_{\pm 0.31}$	$23.3_{\pm 0.46}$	$25.1_{\pm 0.77}$	$8.1_{\pm 1.92}$	$53.8_{\pm 0.72}$	$15.4_{\pm 0.53}$
	After	$14.2_{\pm 3.01}$	$0.2_{\pm 0.31}$	$10.5_{\pm 0.92}$	$26.3_{\pm 1.01}$	$3.6_{\pm 0.63}$	$41.7_{\pm 0.72}$	$11.6_{\pm 0.16}$
Qwen2.5	Before	$46.5_{\pm 0.70}$	$3.1_{\pm 0.31}$	$39.9_{\pm 1.01}$	$25.9_{\pm 3.58}$	$6.9_{\pm 2.23}$	$54.9_{\pm 0.47}$	$23.6_{\pm 0.32}$
	After	$21.7_{\pm 6.23}$	$2.8_{\pm 0.68}$	$12.3_{\pm 1.79}$	$25.9_{\pm 0.77}$	$14.1_{\pm 0.82}$	$40.6_{\pm 0.98}$	$22.9_{\pm 0.70}$
Qwen3	Before	$59.8_{\pm 1.22}$	$3.4_{\pm 0.41}$	$46.5_{\pm 1.86}$	$28.5_{\pm 2.39}$	$12.7_{\pm 1.00}$	$59.1_{\pm 1.23}$	$28.5_{\pm 0.95}$
	After	$8.5_{\pm 1.06}$	$0.5_{\pm 0.27}$	$6.7_{\pm 0.76}$	$29.0_{\pm 2.04}$	$2.0_{\pm 0.47}$	$38.8_{\pm 0.98}$	$11.3_{\pm 0.31}$
MV starting-point controls								
16K extended	Before	$43.7_{\pm 0.93}$	$8.9_{\pm 0.27}$	$31.3_{\pm 1.47}$	$26.9_{\pm 0.29}$	$10.7_{\pm 1.58}$	$46.6_{\pm 0.30}$	$28.0_{\pm 2.73}$
	After	$54.7_{\pm 3.36}$	$13.4_{\pm 0.87}$	$28.8_{\pm 1.91}$	$30.1_{\pm 4.98}$	$18.4_{\pm 0.41}$	$40.9_{\pm 1.14}$	$35.6_{\pm 1.15}$
Tulu3 SFT + DPO	Before	$47.2_{\pm 4.74}$	$6.7_{\pm 2.67}$	$34.6_{\pm 1.06}$	$27.3_{\pm 0.87}$	$8.2_{\pm 0.75}$	$54.6_{\pm 1.38}$	$24.8_{\pm 1.13}$
	After	$46.1_{\pm 1.41}$	$8.6_{\pm 1.49}$	$34.5_{\pm 2.44}$	$27.1_{\pm 1.27}$	$11.5_{\pm 0.51}$	$49.2_{\pm 0.61}$	$25.3_{\pm 0.80}$

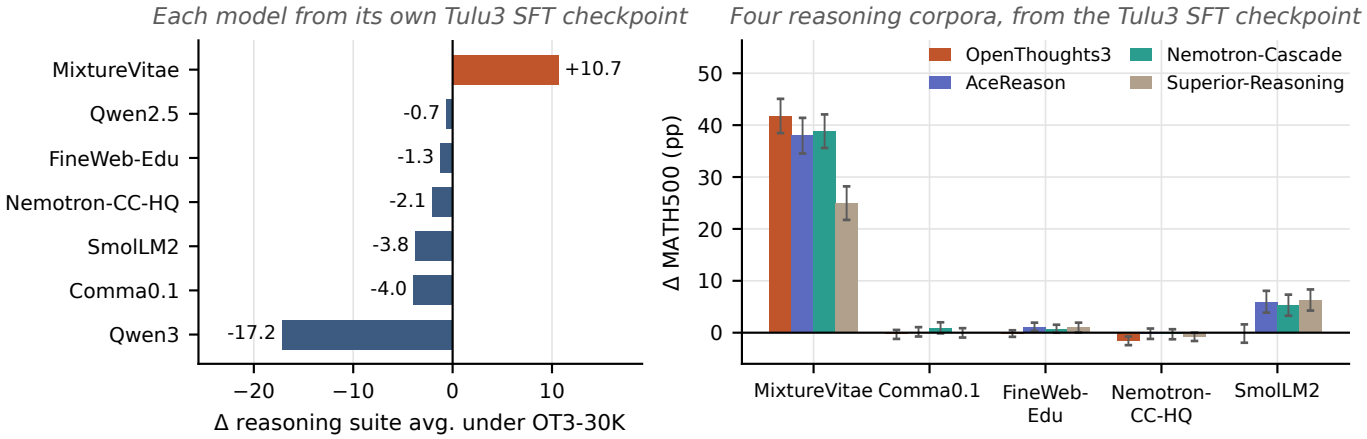
Table 2: **Effect of excluding OpenThoughts3 (OT3) from pretraining.** Two 1.7B models pretrained for 300B tokens receive the same OT3-30K update directly from their 16K extended checkpoints, without Tulu3. Cells show the mean over three sampled evaluations, with SD below. Avg is the unweighted mean of all eleven benchmarks. Evaluation follows Protocol B.

	Mathematics					Code			Science/QA			
Stage	**A24**	**A25**	**AMC**	**M500**	**GSM8K**	**HE**	**LCB**	**MBPP**	**GPQA**	**JEE**	**IFE**	**Avg**
With OT3 in pretraining												
Before	3.3 ±3.33	8.9 ±3.85	25.8 ±12.58	46.5 ±11.29	55.0 ±1.78	43.7 ±0.93	8.9 ±0.27	31.3 ±1.47	26.9 ±0.29	10.7 ±1.58	46.6 ±0.30	28.0 ±2.73
After	8.9 ±5.09	25.6 ±1.92	42.5 ±6.61	69.8 ±1.25	58.8 ±1.52	54.7 ±3.36	13.4 ±0.87	28.8 ±1.91	30.1 ±4.98	18.4 ±0.41	40.9 ±1.14	35.6 ±1.15
Without OT3 in pretraining												
Before	1.1 ±1.92	0.0 ±0.00	12.5 ±4.33	29.8 ±4.33	55.7 ±0.57	36.4 ±1.86	6.6 ±2.60	32.3 ±1.40	26.1 ±0.77	11.0 ±1.89	49.4 ±0.46	23.7 ±0.91
After	2.2 ±1.92	3.3 ±0.00	14.2 ±3.82	38.5 ±1.10	58.5 ±0.22	39.8 ±3.92	5.0 ±1.74	33.5 ±1.33	27.9 ±1.05	10.4 ±1.42	46.1 ±0.46	25.4 ±0.50

Suite-average gain: with OT3, +7.6; without OT3, +1.7 points.

Table 3: **The reasoning gain is not tied to OpenThoughts3.** Change in MATH500 accuracy when each base's Tulu3 SFT checkpoint receives 30K samples from one of four reasoning corpora, three of which are not named sources of any pretraining mixture considered here. Each problem is scored as its mean over three sampled runs. Brackets are 95% paired bootstrap intervals over the same 500 problems. **Bold** marks a significant increase and *italic* a significant decrease after Benjamini–Hochberg correction over the 20 changes. Protocol B (Appendix Table 5).

Base	**OpenThoughts3**	**AceReason**	**Nemotron-Cascade**	**Superior-Reasoning**
MixtureVitae	**+41.8** [+38.5, +45.1]	**+38.0** [+34.5, +41.4]	**+38.9** [+35.6, +42.1]	**+24.9** [+21.7, +28.2]
Comma0.1	−0.3 [−1.2, +0.5]	+0.1 [−0.7, +1.1]	+0.9 [−0.1, +2.0]	+0.0 [−0.9, +0.9]
FineWeb-Edu	−0.2 [−0.8, +0.5]	**+1.1** [+0.3, +1.9]	+0.7 [+0.0, +1.5]	+1.0 [+0.1, +1.9]
Nemotron-CC-HQ	*−1.5* [−2.4, −0.7]	−0.2 [−1.2, +0.8]	−0.3 [−1.3, +0.7]	−0.8 [−1.6, +0.0]
SmolLM2	−0.1 [−1.9, +1.6]	**+5.9** [+3.9, +8.1]	**+5.3** [+3.3, +7.3]	**+6.3** [+4.3, +8.3]



phase2_receptivity.pdf

Table 4: Full grid of reasoning corpora for MixtureVitae: four reasoning corpora applied from three starting checkpoints, with recipe, sample budget, and epoch count held fixed so that only the identity of the reasoning corpus changes. The change in parentheses is against the block's starting checkpoint. Protocol B (Appendix Table 5). Values are mean$_{\pm SD}$ over three sampled runs, and Avg is the mean over the eleven tasks shown.

(a) Mathematics

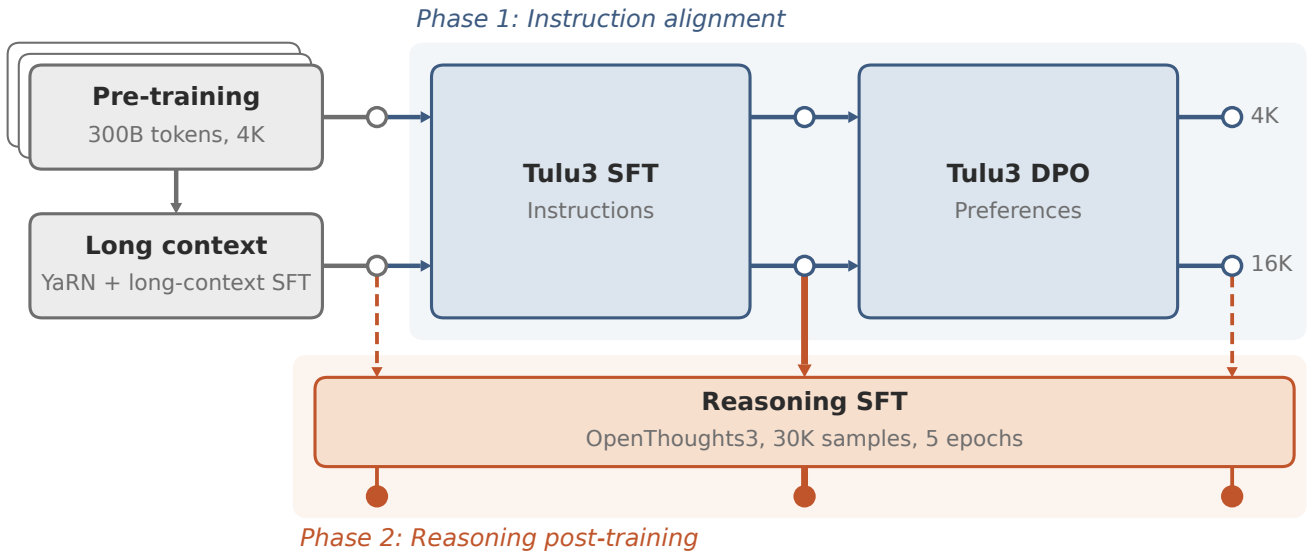
Reasoning corpus	**A24**	**A25**	**AMC**	**M500**	**GSM8K**
Starting from the 16K extended checkpoint					
starting checkpoint	$3.3_{\pm 3.33}$	$8.9_{\pm 3.85}$	$25.8_{\pm 12.58}$	$46.5_{\pm 11.29}$	$55.0_{\pm 1.78}$
OpenThoughts3	$8.9_{\pm 5.09}$	$25.6_{\pm 1.92}$	$42.5_{\pm 6.61}$	$69.8_{\pm 1.25}$	$58.8_{\pm 1.52}$
AceReason	$6.7_{\pm 6.67}$	$13.3_{\pm 0.00}$	$45.0_{\pm 2.50}$	$68.0_{\pm 1.71}$	$59.7_{\pm 0.65}$
Nemotron-Cascade	$11.1_{\pm 8.39}$	$14.4_{\pm 1.92}$	$44.2_{\pm 1.44}$	$67.0_{\pm 1.11}$	$59.9_{\pm 0.31}$
Superior-Reasoning	$4.4_{\pm 1.92}$	$6.7_{\pm 0.00}$	$32.5_{\pm 2.50}$	$51.4_{\pm 0.53}$	$63.0_{\pm 0.74}$
Starting from the Tulu3 SFT checkpoint					
starting checkpoint	$1.1_{\pm 1.92}$	$0.0_{\pm 0.00}$	$8.3_{\pm 3.82}$	$29.2_{\pm 6.41}$	$46.3_{\pm 0.88}$
OpenThoughts3	$11.1_{\pm 3.85}$	$16.7_{\pm 3.33}$	$46.7_{\pm 6.29}$	$71.0_{\pm 1.06}$	$49.0_{\pm 1.75}$
AceReason	$10.0_{\pm 0.00}$	$15.6_{\pm 1.92}$	$45.0_{\pm 4.33}$	$67.2_{\pm 0.60}$	$57.2_{\pm 0.62}$
Nemotron-Cascade	$11.1_{\pm 1.92}$	$16.7_{\pm 3.33}$	$44.2_{\pm 1.44}$	$68.1_{\pm 0.81}$	$72.0_{\pm 0.18}$
Superior-Reasoning	$4.4_{\pm 1.92}$	$7.8_{\pm 5.09}$	$28.3_{\pm 5.20}$	$54.1_{\pm 1.30}$	$57.8_{\pm 0.55}$
Starting from the Tulu3 SFT + DPO checkpoint					
starting checkpoint	$1.1_{\pm 1.92}$	$1.1_{\pm 1.92}$	$14.2_{\pm 6.29}$	$30.1_{\pm 3.31}$	$47.4_{\pm 1.20}$
OpenThoughts3	$0.0_{\pm 0.00}$	$4.4_{\pm 1.92}$	$17.5_{\pm 4.33}$	$34.4_{\pm 1.78}$	$44.4_{\pm 1.52}$
AceReason	$12.2_{\pm 1.92}$	$16.7_{\pm 0.00}$	$40.8_{\pm 6.29}$	$67.1_{\pm 1.33}$	$57.3_{\pm 0.26}$
Nemotron-Cascade	$7.8_{\pm 1.92}$	$18.9_{\pm 1.92}$	$43.3_{\pm 7.64}$	$68.7_{\pm 1.51}$	$71.8_{\pm 0.36}$
Superior-Reasoning	$6.7_{\pm 3.33}$	$6.7_{\pm 0.00}$	$29.2_{\pm 3.82}$	$53.5_{\pm 1.81}$	$57.6_{\pm 0.38}$

(b) Code, science and instruction following, and the average

Reasoning corpus	**HE**	**LCB**	**MBPP**	**GPQA**	**JEE**	**IFE**	**Avg**
Starting from the 16K extended checkpoint							
starting checkpoint	$43.7_{\pm 0.93}$	$8.9_{\pm 0.27}$	$31.3_{\pm 1.47}$	$26.9_{\pm 0.29}$	$10.7_{\pm 1.58}$	$46.6_{\pm 0.30}$	$28.0_{\pm 2.73}$
OpenThoughts3	$54.7_{\pm 3.36}$	$13.4_{\pm 0.87}$	$28.8_{\pm 1.91}$	$30.1_{\pm 4.98}$	$18.4_{\pm 0.41}$	$40.9_{\pm 1.14}$	$35.6_{\pm 1.15}$ (+7.6)
AceReason	$58.3_{\pm 0.70}$	$13.8_{\pm 0.27}$	$37.5_{\pm 1.89}$	$29.8_{\pm 3.15}$	$20.6_{\pm 0.88}$	$40.2_{\pm 1.24}$	$35.7_{\pm 0.72}$ (+7.7)
Nemotron-Cascade	$54.7_{\pm 2.82}$	$12.6_{\pm 0.78}$	$37.7_{\pm 1.63}$	$28.6_{\pm 1.27}$	$20.6_{\pm 0.60}$	$43.1_{\pm 0.87}$	$35.8_{\pm 0.68}$ (+7.8)
Superior-Reasoning	$31.5_{\pm 5.67}$	$8.6_{\pm 0.41}$	$20.6_{\pm 1.06}$	$25.9_{\pm 1.62}$	$10.4_{\pm 1.03}$	$41.6_{\pm 0.61}$	$27.0_{\pm 0.34}$ (−1.0)
Starting from the Tulu3 SFT checkpoint							
starting checkpoint	$44.9_{\pm 1.76}$	$7.0_{\pm 2.26}$	$31.9_{\pm 0.46}$	$27.4_{\pm 0.58}$	$7.8_{\pm 1.33}$	$54.0_{\pm 0.15}$	$23.4_{\pm 0.31}$
OpenThoughts3	$51.0_{\pm 4.97}$	$11.5_{\pm 0.87}$	$28.5_{\pm 3.29}$	$28.8_{\pm 1.75}$	$18.8_{\pm 1.34}$	$42.1_{\pm 0.20}$	$34.1_{\pm 0.93}$ (+10.7)
AceReason	$54.7_{\pm 1.41}$	$14.2_{\pm 1.10}$	$36.1_{\pm 0.99}$	$26.1_{\pm 1.17}$	$20.9_{\pm 1.05}$	$41.6_{\pm 0.34}$	$35.3_{\pm 0.47}$ (+11.9)
Nemotron-Cascade	$55.1_{\pm 0.93}$	$13.6_{\pm 1.76}$	$37.2_{\pm 1.59}$	$28.1_{\pm 0.58}$	$20.9_{\pm 1.14}$	$44.1_{\pm 0.98}$	$37.4_{\pm 0.34}$ (+13.9)
Superior-Reasoning	$29.5_{\pm 1.86}$	$7.8_{\pm 0.56}$	$21.8_{\pm 0.35}$	$25.3_{\pm 2.31}$	$10.2_{\pm 1.06}$	$41.2_{\pm 0.61}$	$26.2_{\pm 0.48}$ (+2.7)
Starting from the Tulu3 SFT + DPO checkpoint							
starting checkpoint	$47.2_{\pm 4.74}$	$6.7_{\pm 2.67}$	$34.6_{\pm 1.06}$	$27.3_{\pm 0.87}$	$8.2_{\pm 0.75}$	$54.6_{\pm 1.38}$	$24.8_{\pm 1.13}$
OpenThoughts3	$46.1_{\pm 1.41}$	$8.6_{\pm 1.49}$	$34.5_{\pm 2.44}$	$27.1_{\pm 1.27}$	$11.5_{\pm 0.51}$	$49.2_{\pm 0.61}$	$25.3_{\pm 0.80}$ (+0.5)
AceReason	$52.8_{\pm 7.74}$	$14.8_{\pm 0.87}$	$36.7_{\pm 0.50}$	$29.0_{\pm 2.59}$	$20.7_{\pm 0.86}$	$42.1_{\pm 1.66}$	$35.5_{\pm 0.71}$ (+10.7)
Nemotron-Cascade	$52.8_{\pm 4.06}$	$14.4_{\pm 0.72}$	$37.7_{\pm 1.62}$	$28.6_{\pm 0.77}$	$22.2_{\pm 0.44}$	$43.9_{\pm 1.43}$	$37.3_{\pm 1.03}$ (+12.5)
Superior-Reasoning	$29.7_{\pm 1.41}$	$6.2_{\pm 0.00}$	$22.9_{\pm 1.33}$	$24.6_{\pm 2.28}$	$10.1_{\pm 1.77}$	$41.2_{\pm 0.15}$	$26.2_{\pm 1.18}$ (+1.4)

Table 5: Protocol B, the reasoning post-training protocol. Used for every OpenThoughts3-30K experiment, namely the endpoints of Table **??**, the pretraining ablation of Table 2, and the grid of reasoning corpora of Tables 3 and 4. Every task is sampled at temperature 0.7, with one answer per question in each of three runs, and results are the mean and SD over the runs.

Abbr.	Task	Metric	Questions	Max new toks
A24	AIME24	Accuracy	30	32,000
A25	AIME25	Accuracy	30	32,000
AMC	AMC23	Accuracy	40	32,000
M500	MATH500	Accuracy	500	32,000
GSM8K	GSM8K	Exact match, flexible answer extraction	1,319	16,000
HE	HumanEval	Pass@1, Python problems	164	16,000
LCB	LiveCodeBench	Accuracy	369	16,000
MBPP	MBPP	Pass@1	500	16,000
GPQA	GPQA-Diamond	Accuracy	198	32,000
JEE	JEEBench	Accuracy	515	32,000
IFE	IFEval	Instruction-level accuracy, strict	500	16,000



figure_pipeline.pdf